

# Proyección estereográfica

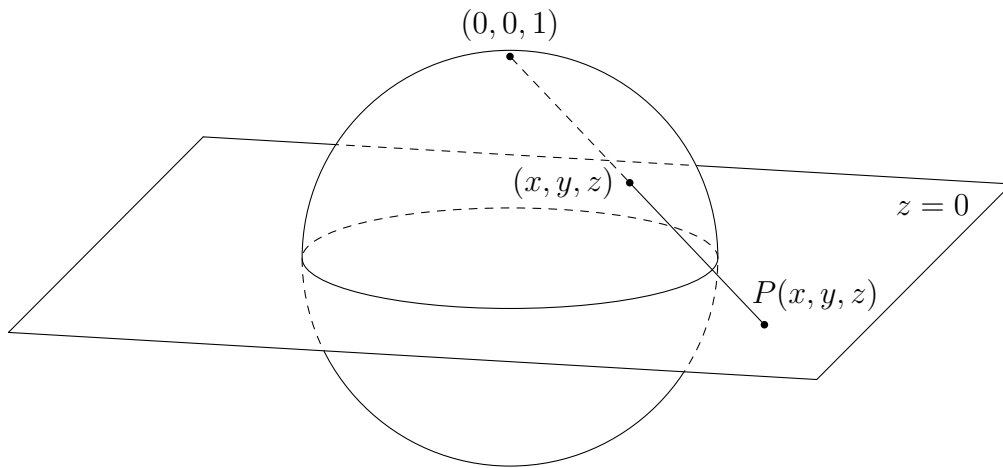
**Definición 1 (Proyección estereográfica).** Sea  $\mathbb{S}^2 = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$  la esfera unidad en  $\mathbb{R}^3$ ,  $P: \mathbb{S}^2 \setminus \{(0, 0, 1)\} \rightarrow \mathbb{R}^2$  es la proyección estereográfica

$$\iff \forall (x, y, z) \in \mathbb{S}^2 \setminus \{(0, 0, 1)\} : P(x, y, z) = \left( \frac{x}{1-z}, \frac{y}{1-z} \right).$$

La función inversa queda:

$$P^{-1}: \mathbb{R}^2 \rightarrow \mathbb{S}^2 \setminus \{(0, 0, 1)\}$$

$$(a, b) \mapsto \frac{1}{a^2 + b^2 + 1} (2a, 2b, a^2 + b^2 - 1)$$



Referenciado en

- plano-complejo-extendido

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